

Polarography:-

→ Polarography is the type of Voltammetry. This technique is used for qualitative and quantitative analysis.

Construction of polarographic cell:-

→ Polarographic cell consists of three electrodes.

- ✓ Working electrode (we used dropping mercury electrode).
- ✓ Reference electrode (Saturated calomel electrode is used).
- ✓ Mercury pool is act as a anode.

→ DC current is applied.

→ The diameter of capillary is very important (the specific time for mercury drop will be 2 to 6 second).

→ The length of Hg. pool is about 50 cm.

→ The length of capillary is about 10 cm.

→ A potentiometer is used to vary the voltage.

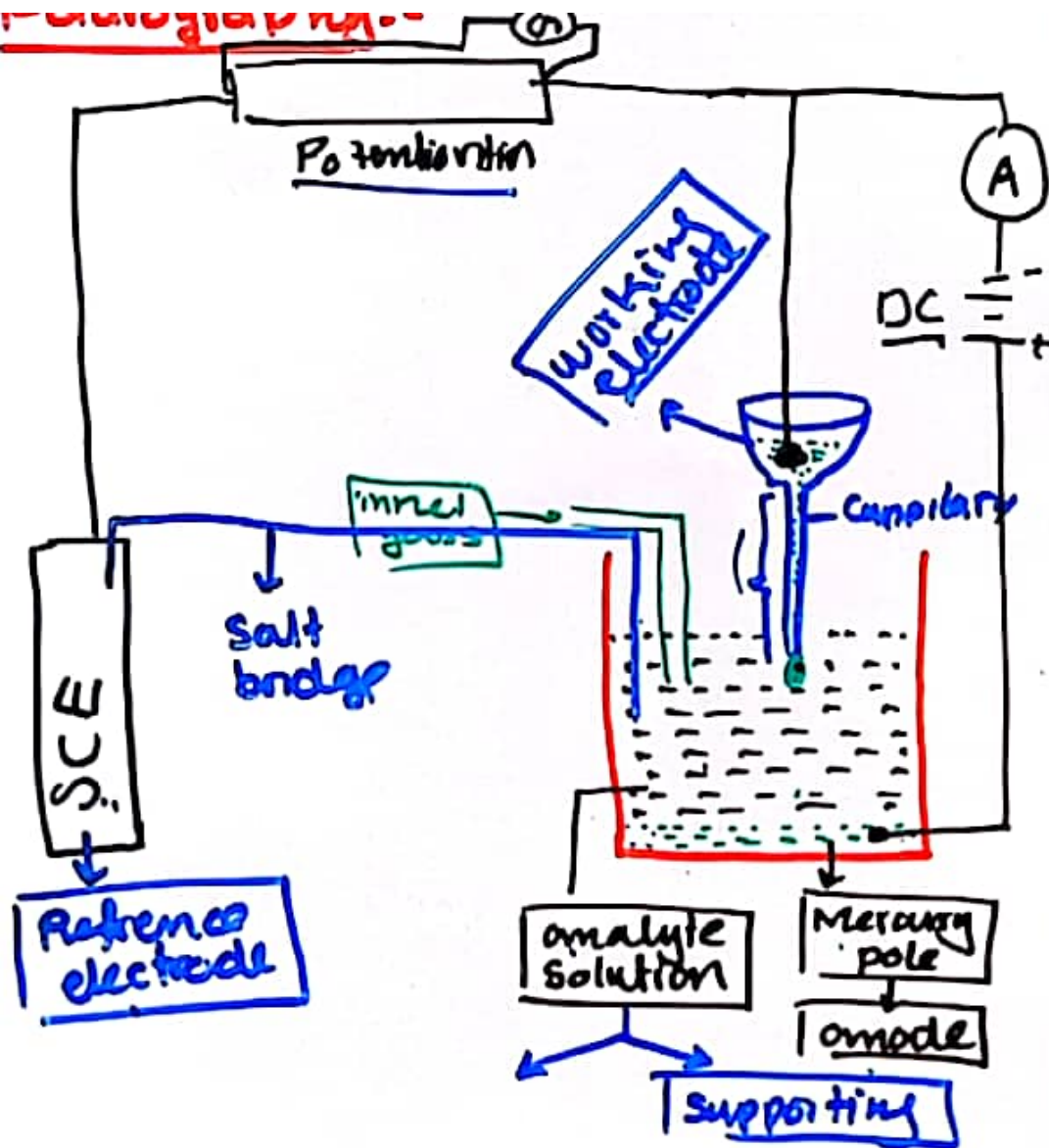
→ Ammeter and galvanometer is used to determine the voltage and current.

→ A supporting electrolyte is also used, which decrease the migration current.

→ An inert gas is used to remove the dissolved oxygen from the solution.

→ A salt bridge is used to connect the analyte solution & reference electrode.





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Polarographic technique:-

Working electrode:-

→ Dropping mercury electrode is used in polarographic technique.

Advantages:-

- Hg yield reproducible current & potential data.
- Analysis of acid is possible because Hg does not react chemically with.
- The vast majority of reduction reaction can be induced by using DME.
- There is high over potential for H^+ ions reduction. at the mercury surface, therefore H^+ ions does not interfere

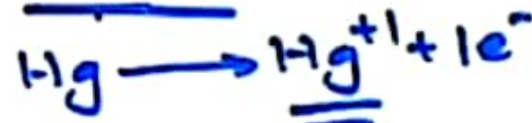


Disadvantages:-

→ A Hg-electrode is not very useful for performing oxidation. because Hg is too easily oxidized.



→ In a non-complexing medium, Hg oxidize near +0.26V.



→ Hg is poisonous and volatile.

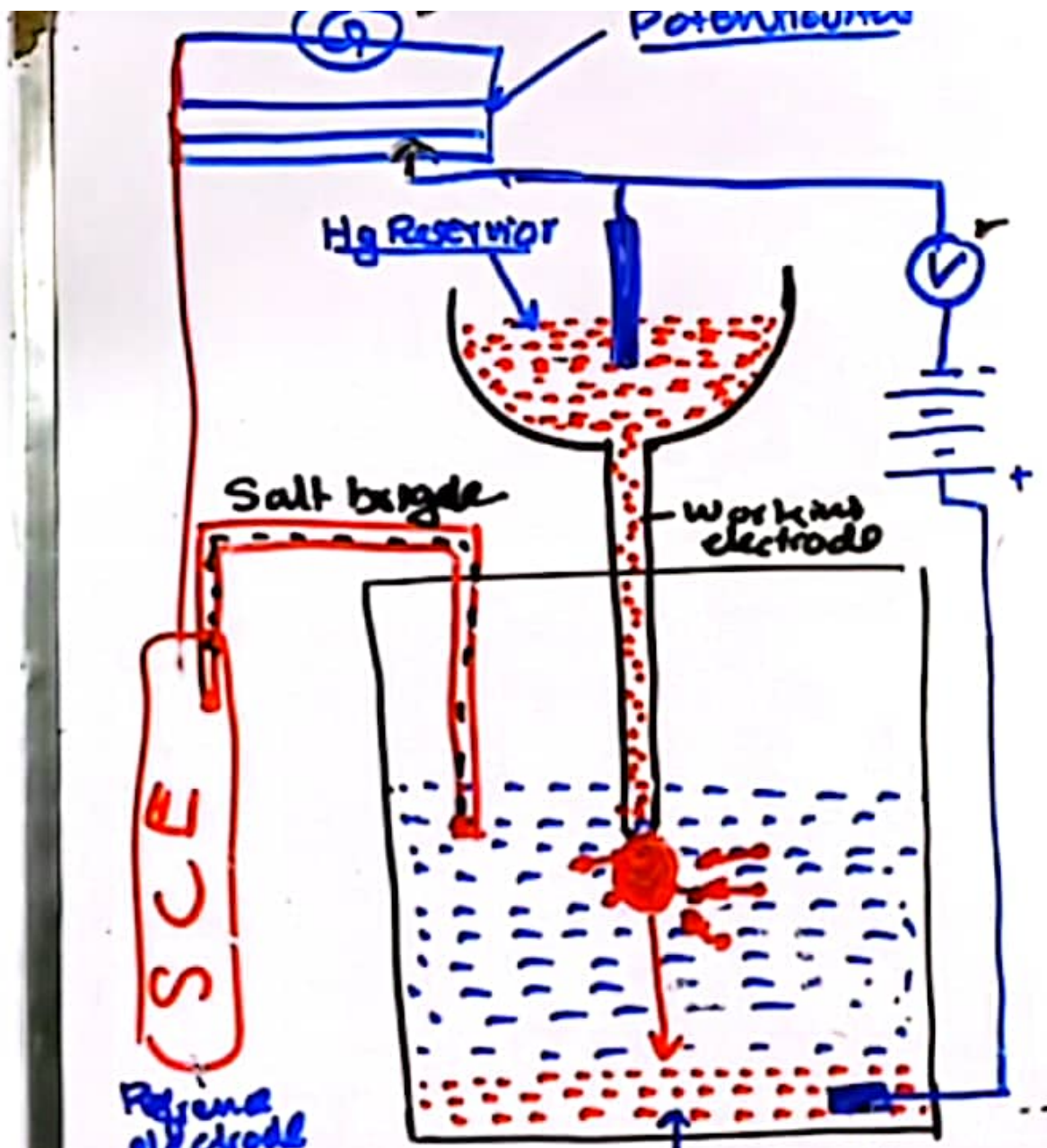
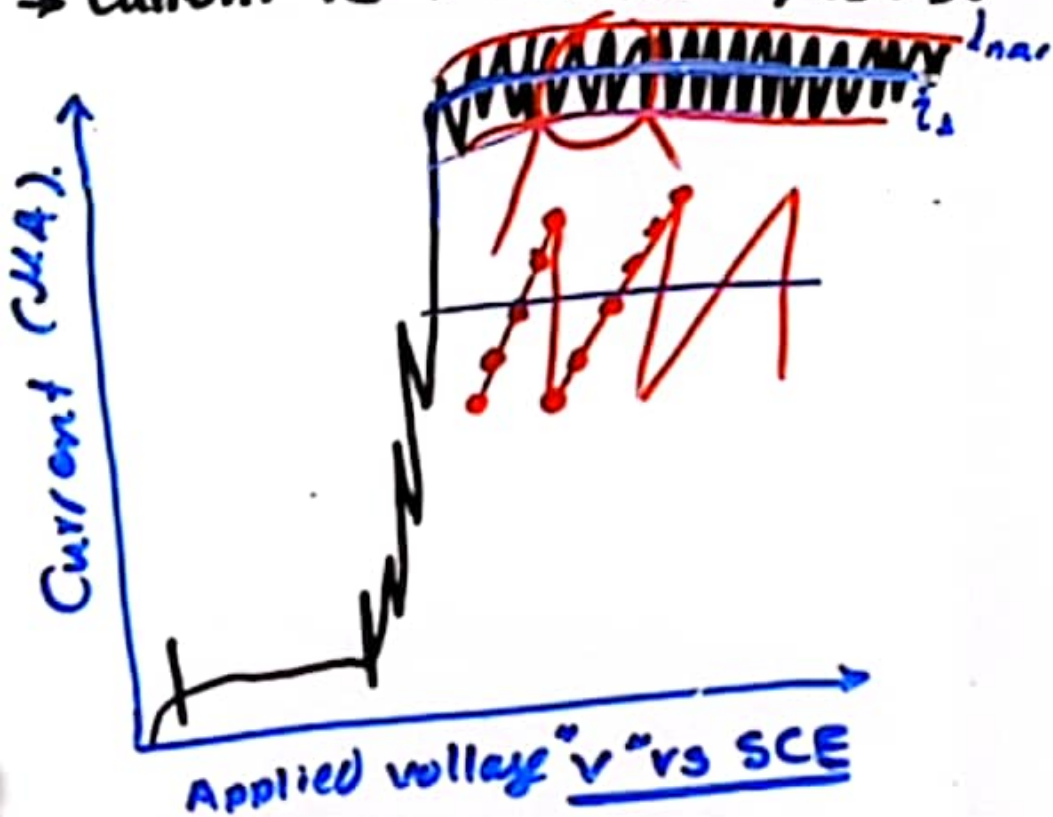
→ To remove residual Hg, sprinkle elemental Zn-powder on the surface, and dampen with 5% Aqueous H_2SO_4 .

Polarography:-

↳ Qualitative
Quantitative Analysis.

Polarogram:- The graph b/w Voltage and current produce in polarography is called....

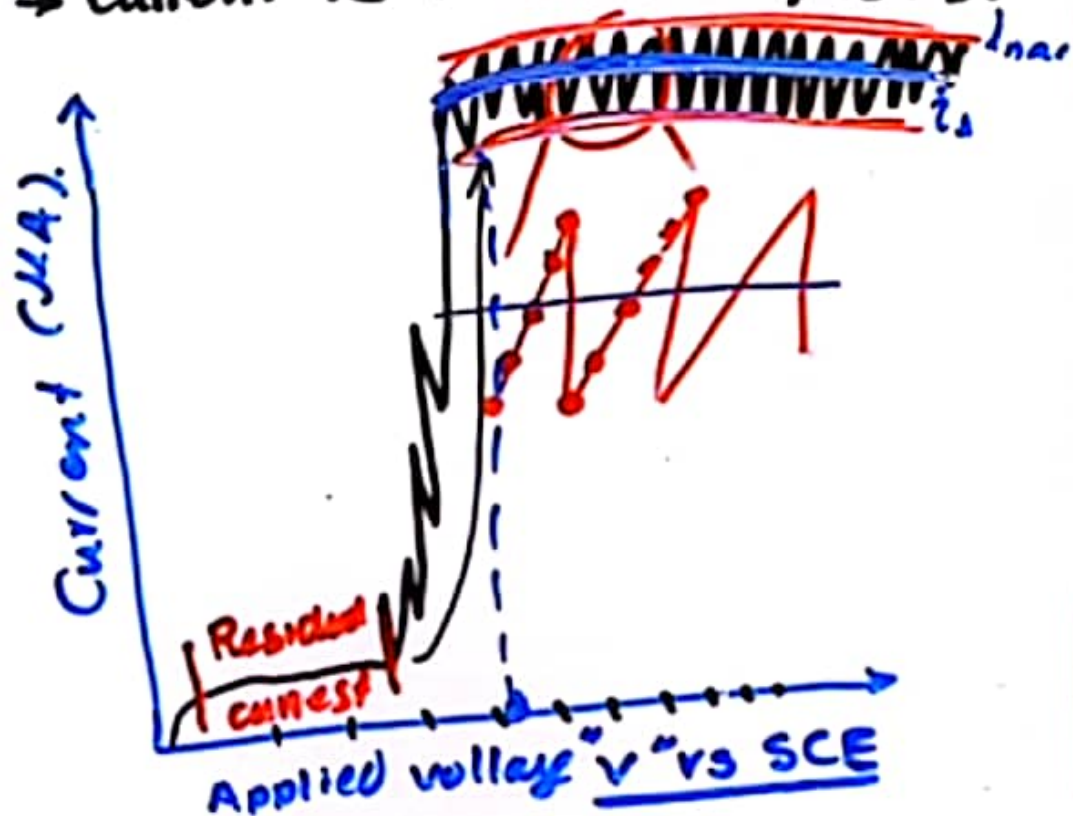
- Voltage is taken along x-axis
- Current is taken on y-axis.



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Residual current:- Residual current is produce due to impurities.

- ✓ Faradaic current
- ✓ Non-Faradaic current.

Diffusion Current:-

The current produce due to the oxidation or reduction of analyte.



Limiting Current:-

Limiting Current is the maximum current obtain in polarography. Diffusion current can be calculate by using Ilkovic equation.

$$I_d (\text{max}) = 706 n D^{1/2} m^{2/3} t^{1/2} C$$

$$I_d (\text{ave}) = 607 n D^{1/2} m^{2/3} t^{1/2} C$$

Diffusion current $n = \text{no. of } e^-$ Diffusion Coefficient Rate Flow of Hg $t = \text{life time of drop}$

Ilkovic Equation:-

Ilkovic Equation is described the relationship b/w diffusion & concentration of analyte.

$$I_d(\text{max}) = 706nD^{1/2}m^{2/3}t^{1/6}C$$

$$I_d(\text{ave}) = 607nD^{1/2}m^{2/3}t^{1/6}C$$

where "n"
 I_d = Diffusion current, measured at the top of oscillation in the figure (uA)

$\therefore n$ = no. of electron per molecule involve in oxidation or reduction of the electro-active specie

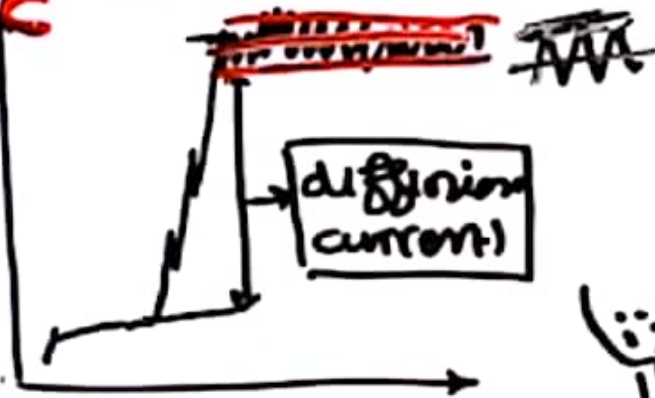


$\therefore D$ = Diffusion coefficient of electroactive (M^2/s) specie.

$\therefore m$ = rate of mass flow of Hg in Capillary (mg/s)

$\therefore t$ = life time of Hg-drop. ✓

$\therefore C$ = Concentration of analyte.



$$I_d = kC$$

$$I_d \propto C$$

$\therefore$ The diffusion current is directly proportional to the concentration of analyte.

Polarography:-

Factors effecting on diffusion Current:-

Concentration:-

$$I_d = KC \quad \therefore I_d = \text{diffusion current}$$

$$\therefore C = \text{concentration}$$

Diffusion current is directly proportional to the concentration of analyte ion.

Temperature:-

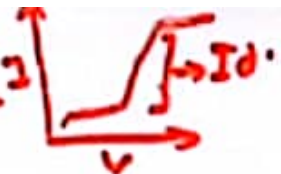
$I_d = 706 n D^{1/2} \frac{m}{t^{1/2}}$

Temperature effect the viscosity of Hg. when "T" increase the viscosity decrease. Due to decrease in viscosity rate of flow increase and "I_d" change.

→ The whole experiment is performed at constant temperature (25°C).

Life time of Mercury drop:-

when life time of Hg drop is smaller, then small amount of analyte deposit and small diffusion current is produce.



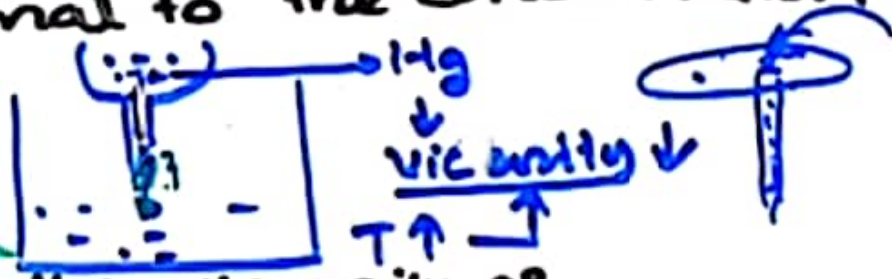
Capillary Constant:-

→ m & $t^{1/2}$ is called capillary const.

m = rate of mass flow of Hg in capillary (mg/s).

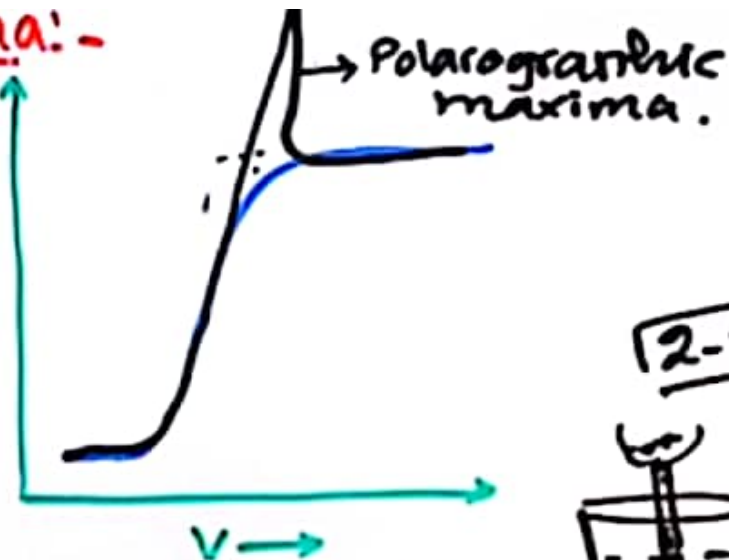
t = Life time of Hg drop in second.

$$\left. \begin{aligned} I_d &\propto m \\ I_d &\propto t \end{aligned} \right\}$$



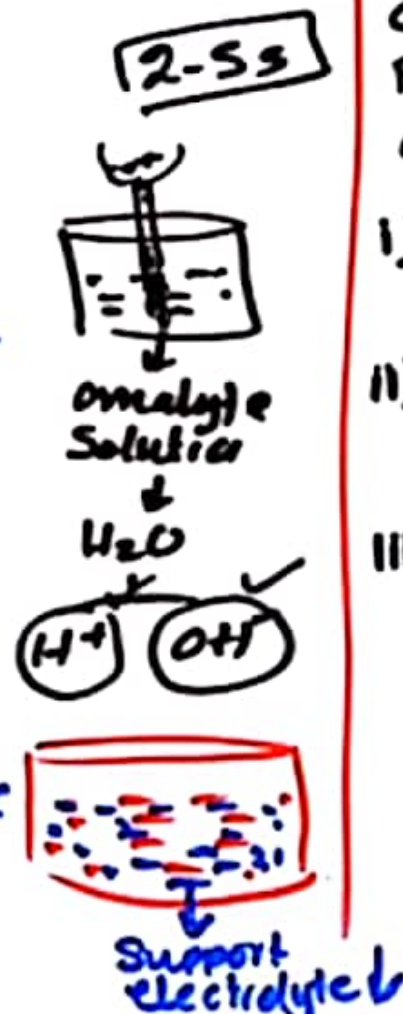
Polarographic (V) Maxima:-

→ The abnormal increase of limiting current is called Polarographic Maxima.



Cause of Polarographic Maxima:-

- The exact reason for polarographic maxima is still unknown.
- It may be caused during the catalytic discharge H^+ & OH^- .
- If the Hg-flow is too high then polarographic maxima is observed.
- The ^{low} concentration of supporting electrolyte may cause polarographic maxima.



Removal of Polarographic Maxima:

Reasonable amount of surfactant can remove polarographic Maxima.

- I) Triton X-100 (non-ionic detergent)
- II) Rabon (0.009 to 0.0049%)
- III) Gelatin

Applications of Polarography:-

- It is used for qualitative and Quantitative analysis.
- It is used for the analysis of trace metals in the alloys.
 - ↳ Element
 - ↳ $E_{1/2}$
- Used for environmental analysis.
- It is used in food stuff, beverage, body fluid, clinical analysis.

$$\boxed{Id} \quad \boxed{Id \propto C}$$

Biological system:-

It is used to determine the concentration of vitamins, alkaloids, hormones, terpenoids & so on.

Medical field:-

- analyze natural coloring substance in drugs.
- Pharmaceutical preparation.
- Determination of pesticides or herbicides.
- Used to determine the structure of organic compound.